Interdisciplinary Project Documentation

IP05

Development Group 2

Course: Interdisciplinary Project - Social Responsibility

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1 Overview

This Open Terms Archive Frontend project was developed to provide a user-friendly platform that helps users understand how Terms of Service documents change over time. Digital service providers frequently update their legal documents, which makes it difficult for users to identify what has changed and how those changes may affect them. To address this challenge, the project offers a comparison mechanism that allows users to compare multiple versions of Terms of Service documents and clearly identify updates. The platform is part of the broader Open Terms Archive initiative [1], which aims to increase transparency around online terms and policies.
The platform highlights differences between document versions using color-coded indicators. Newly added or modified content in updated versions is highlighted in green, while content that has been removed or altered from previous versions is highlighted in red. This visual representation simplifies the review process and significantly improves user comprehension by allowing users to focus directly on relevant changes without reading entire documents.
In addition, the platform integrates multiple repositories, including a dedicated collection of German companies, sourced from Open Terms Archive repositories [2]. By combining clear visual change tracking with access to structured and reliable data sources, the frontend delivers an accessible and transparent solution that supports informed decision-making and aligns with the mission of Open Terms Archive.
This project was developed as part of the interdisciplinary project IP05 and was supervised by Prof. Matteo Grosse-Kampmann and Omed Abid, focusing on the design and implementation of a frontend solution for Open Terms Archive

1.1 Project Scope

This project is a part of a broader interdisciplinary initiative associated with the Open Terms Archive. The focus of the work is to design, build and document the frontend.
Accordingly, the focus of this work lies in the presentation and interaction layer which includes rendering document versions, enabling visual comparison, supporting multilingual access, and ensuring a clear, responsive, and user-friendly interface. Defining these boundaries ensures a clear separation of responsibilities and accurately reflects the frontend's role within the Open Terms Archive ecosystem.

2 Problem Statement

Terms of Service documents play a crucial role in defining the legal relationship between digital service providers and users.
However, these documents are often lengthy and are frequently updated without clear indication what has changed in the latest version. As a result, users typically accept updated terms without fully understanding the modifications or their potential impact.
While Open Terms Archive maintains comprehensive repositories of Terms of Service documents and their historical versions, the raw repository format is not easily accessible for non technical users. Users face difficulties in navigating between versions, identifying specific changes, and interpreting updates across different time periods.
This project addresses these challenges by developing a frontend application that transforms complex and versioned text data into an accessible and visually clear format for users.

3 Objectives

The primary objective of this project is to improve transparency and accessibility of Terms of Service documents by enabling users to clearly identify and understand changes made over time. The project aims to bridge the gap between complex legal document and user. The specific objectives of the project are as follows:

- To design and implement a frontend interface that allows users to browse and access multiple versions of Terms of Service documents.
- To provide a clear version comparison mechanism that visually highlights additions, modifications, and removals between document versions.
- To apply color-coded indicators to represent changes, improving readability and reducing cognitive effort for users.
- To integrate multiple repositories, including a dedicated repository containing German companies' Terms of Service.
- To develop a responsive and consistent user interface using modern frontend technologies and design principles.
- To support multilingual content in English, German, and French to enhance accessibility for a wider audience.
- To ensure the frontend aligns with the transparency and accountability goals of Open Terms Archive.

These objectives guided the design, development, and evaluation of the frontend application throughout the project lifecycle.

4 Architecture

The project is designed as a component-based application in a clear and structured manner. The system architecture fetches data from Open Terms Archive repositories and the German Repository (Made by our parter Developer team) . It then processes it for user-friendly presentation. The application architecture consists of three main layers

- The first layer is the data access layer, which is responsible for fetching Terms of Service documents and their different versions from the Open Terms Archive repositories, including the repository containing German companies. This layer handles communication with external data sources and prepares the raw data for further processing.
- The second layer is the processing and comparison layer. In this layer, document versions are analyzed and compared to identify textual differences between successive versions. The comparison logic determines which content has been added, modified, or removed. These differences are then categorized and passed to the presentation layer for visual highlighting.
- The third layer is the presentation layer, which renders the processed data to the user through a responsive and interactive user interface. This layer is responsible for displaying document versions, applying color-coded highlights to indicate changes, and supporting multilingual content.

5 Project Structure

The project is structured in a manner in which user interface components, shared logic, data access, and translation resources are organized separately. This separation allows each part of the application to be developed and updated independently.

The structure made it easier for the team to work on different parts of the frontend at the same time and to make changes without affecting unrelated components. It also allows new repositories, interface adjustments, or additional languages to be added without restructuring the existing code. This separation of concerns helps keep interface components lightweight and focused on presentation, while shared logic and data handling are managed independently. As a result, the frontend remains easier to understand, debug, and extend during development.

Overall, the project structure supports a clear and manageable frontend codebase appropriate for the scope of this project.

6 Technologies Used

The Open Terms Archive Frontend was developed using modern web technologies to support a component-based user interface and dynamic rendering of versioned documents.

Chakra UI[3] was used to build a responsive and accessible interface, ensuring consistent layout, readable legal text, and clear visual highlighting of changes across devices.

React i18next[4] was used to handle translation and internationalization, allowing the interface to be displayed in multiple languages.

Diff library[5] was used to compare different versions of documents and highlight additions and deletions.

TypeScript[6] was used to enforce static typing and clear data models, reducing runtime errors and improving code maintainability when handling structured document data, version comparisons, and multilingual content.

React[7] was used to implement reusable interface components for displaying Terms of Service, comparing document versions, and updating the view dynamically based on user interaction.

React Markdown[8] was used to correctly display documents and render content written in markdown format.

Additional libraries supported internationalization and data handling, while GitLab was used for version control and collaborative development throughout the project.

7 Project Journey and Sprint Execution

The development of the Open Terms Archive Frontend followed an agile, sprint-based workflow. Each sprint was carefully planned to ensure structured progress, team collaboration, and timely delivery of features. The team was divided into two sub-groups:

Dev Team 1 – Backend: Responsible for data integration, APIs, and backend support.

Dev Team 2 – Frontend: Focused on UI design, visual elements, and frontend functionality.

Sprint 1: Planning, Blueprint, and Project Structure

Duration: *2 October 2025 – 1 November 2025*

First Half (2 Oct – 16 Oct 2025)

The sprint began with discussions on how to approach the project and determine the best starting point. The team collaboratively created a project blueprint, defining goals, scope, and initial architecture. Responsibilities were divided, and the team was organized into two dedicated development teams to enable parallel progress and efficient workflow.

Second Half (17 Oct - 31 Oct 2025)

In the latter half of the sprint, the team initialized the project and created its structure, setting up the foundational framework that would support further development. This included configuring the development environment, establishing project directories, and implementing initial dependencies for both frontend and backend teams.

Sprint 2: Document Viewing and Version Comparison

Duration: *1 Nov 2025 – 30 November 2025*

Sprint 2 focused on implementing the core functionalities of the platform. The team developed the document viewing feature, allowing users to access Terms of Service documents efficiently. Additionally, a version comparison mechanism was implemented, enabling users to identify and analyze changes between multiple document versions. These features formed the foundation for later enhancements, such as visual change highlighting and structured change summaries.

Sprint 3: Multilingual Support, UI Refinement, and Finalization

Duration: *1 December 2025 – 14 January 2025*

First Half (1–22 December 2025)

The team implemented multilingual support, enabling the platform to serve users in English, German, and French. Work also focused on homepage development and UI refinement, improving navigation, layout, and responsiveness. Concurrently, ongoing bug fixes were applied to ensure stability and usability across devices.

Second Half (5 Jan –14 Jan 2025)

The final phase was dedicated to documentation, project demonstration, and user feedback. Comprehensive documentation was prepared, detailing system functionality, design decisions, and implementation processes. A demo was conducted to showcase the platform's features, during which symbols provided by the psychology team were integrated to enhance usability and visual clarity.
A survey was then conducted to gather user feedback and evaluate the user experience. Based on these reviews, the application underwent final polishing, resulting in a fully functional, user-friendly, and visually coherent frontend solution.

8 Team Contribution

Aarij Muhammad Jan: As the team lead, was responsible for task division and coordination within the team. Managed meetings with other project teams and delivered project presentations. Developed the View Document functionality in the frontend. In addition, handled project documentation, maintained development workflows, and conducted the user survey to evaluate the frontend usability.

Naveen Kumar: Implemented the translation functionality that enables multilingual support within the frontend. Integrated the German repository into the system and ensured correct data handling. Also acted as the GitLab maintainer, managing branches, merges, and overall repository organization to support smooth team collaboration. Created a demo in which symbols provided by the psychology team were added.

Hadees Ayub Implemented character escaping for correct handling of special characters. Built a Markdown-to-HTML conversion pipeline for accurate document rendering. Assisted with bug fixes, documentation, and the migration of project materials into Overleaf.

Syed Mohammad Hasan Naqvi: Designed and developed the frontend of the website, focusing on layout and visual structure. Carried out multiple UI refinements across the frontend to improve consistency, readability, and overall user experience.

Iliya Pleshakov: Responsible for the initial project setup and configuration of the frontend environment. Implemented the document version comparison feature, which allows users to view differences between document versions. Additionally, identified and resolved several bugs to improve system stability.

9 Features

The frontend provides a focused set of features designed to improve the readability and usability of Terms of Service documents Which enables users to compare different versions of a document and understand changes efficiently.

A key feature of the platform is **visual change highlighting**. In the side-by-side document comparison view, two versions of a document are displayed simultaneously, with changes clearly marked (newly added or modified content is highlighted in green, while removed or altered content is highlighted in red.)

In addition, the platform provides a **change summary feature**, which presents only the modifications made, displayed line by line. This allows users to focus exclusively on the changes without reviewing the full documents.

The frontend also emphasizes **text readability and interface clarity**, presenting content in a structured and easy-to-read format.

In addition, the application supports **multilingual access** in English, German, and French and offers a **responsive user interface** that adapts to different devices, ensuring consistent usability across platforms.

10 User Experience Evaluation

A user experience survey was conducted to assess the clarity and usability of the User Interface. Most participants reported that the website appeared clear at first glance, indicating that the layout and navigation are generally intuitive for first time users. A smaller number of participants described the interface as moderately clear suggesting room for minor improvements.
Feedback showed that the visual colour-based highlighting effectively supports understanding of changes between document versions. Users were able to identify updates quickly without reviewing entire documents, aligning with the project's goal.
Qualitative responses highlighted potential enhancements, including clearer section naming, improved spacing and alignment, smoother transitions, and optional features such as a glossary or AI-assisted support. Overall, the survey results indicate a positive user experience while providing useful guidance for future interface refinements.

11 Future Improvements

While the Open Terms Archive Frontend successfully meets its core objectives, several opportunities for future enhancement have been identified to further improve usability, performance, and analytical capabilities.

One potential improvement is the automation of standardized symbol assignment by using Large Language Models (LLMs) or similar Natural Language Processing (NLP) techniques to detect privacy-related content. This would allow the application to automatically identify sensitive topics such as camera access, location tracking, payment information, or personal data usage and annotate them with appropriate standardized symbols provided by the Psychology team.

To improve usability, a global search feature could be introduced that allows users to search for companies across all available repositories, rather than being limited to individual data sources. This would enable faster access to relevant Terms of Service documents and comparison.

The ability to export comparison results in formats such as PDF would further support documentation, reporting, and offline analysis.

From an architectural perspective, introducing URL-based routing would improve navigation and share ability, allowing users to directly access specific companies, repositories, or comparison views via dedicated URLs. Integrating React Query for backend data handling could significantly

enhance data fetching, caching, and error management, leading to improved performance and a more responsive user experience.

Finally, adding a dark mode option would enhance accessibility and user comfort, while maintaining consistency with modern user interface standards.

12 Conclusion

The project successfully delivers a user-centric solution for improving the transparency and accessibility of Terms of Service documents.
By transforming complex and frequently updated legal texts into a clear, visually comparable format, the platform addresses a significant usability gap in how digital contractual information is presented to users.
Through features such as version comparison, color-coded change highlighting, multilingual support, and integration of multiple repositories, the frontend aligns closely with the mission of Open Terms Archive to promote informed decision-making and accountability in digital services. The use of modern frontend technologies and a modular architecture further ensures maintainability and scalability of the application.
In addition to its technical contributions, the project demonstrates effective teamwork, project management, and communication. Overall, the project represents a meaningful academic and practical contribution, combining technical implementation with social responsibility by empowering users to better understand the terms that govern their digital interactions.

Glossary

Chakra UI
A React-based component library used to build accessible, responsive, and visually consistent user interfaces.

Component-Based Architecture
A software design approach in which the user interface is built from reusable, independent components.

Frontend
The part of a web application that users directly interact with, responsible for presentation, interaction, and user experience.

Multilingual Support
The capability of an application to present content and interface elements in multiple languages.

Open Terms Archive (OTA)
An open-source initiative that collects, archives, and tracks changes in Terms of Service, Privacy Policies, and similar legal documents over time.

Privacy Policy
A legal document that explains how a service provider collects, uses, and protects personal data in accordance with data p
rotection laws.

Repository
A structured storage location for project files or datasets, often managed using version control systems such as Git.

Responsive Design
A design approach that ensures a web application adapts effectively to different screen sizes and devices.

Terms of Service (ToS)
A legal agreement between a service provider and its users outlining the rules, responsibilities, and conditions of using a service.

TypeScript
A statically typed programming language that builds on JavaScript and improves code reliability and maintainability.

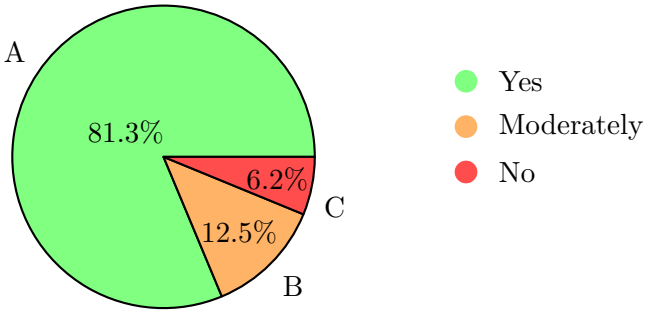
Version Comparison
The process of comparing different versions of a document to identify changes over time.

Visual Change Highlighting
A method of representing document changes using visual cues, such as color coding, to make updates easier to identify.

Appendices

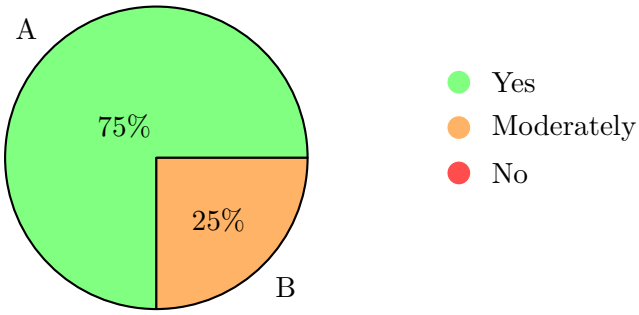
QUESTION 1: How clear as the website at first glance?

16 responses



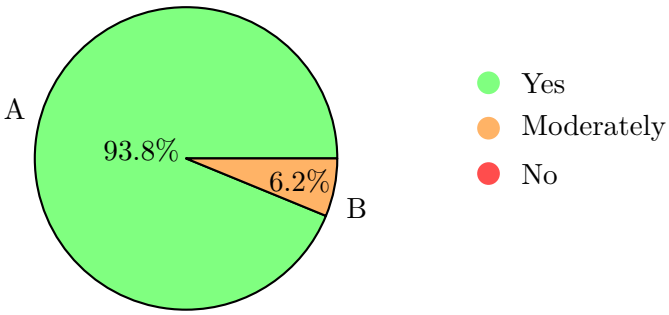
QUESTION 2: Did you find the symbols relevant and helpful?

16 responses



QUESTION 3: Did the different colours made it easier for you to understand what changed?

16 responses



QUESTION 4: what do you like the most about the website?

16 responses

- The UI design
- Simplicity
- The concept
- I liked the grouping of particular websites like the dating section or the generative AI section
- Organization
- Finding specific info, like the German policy of ChatGPT was super easy. Translation of the website.
- Visualisation
- Professional look and easy access
- Everything
- The simplicity
- Its really easy to see what changes are done every year. So the informations are good to see
- Clean design, all information is accessible without looking through a lot of menus
- It's a clean design
- Explanation of what archives are for, low entry level for users
- the simplicity of the UI and the idea of the project
- The UI and ease of access

QUESTION 5: What could be improved?

16 responses

- No need of changes
- Nothing
- User interface
- I cannot name anything right now that could be improved
- Some section names could be clearer. Some words were added instead of replaced, but they appear in the removed section when comparing the documents.
- The tiles on the starting page could be sorted. Like countries in the first line, types in the second and firms in the third or something like that. Maybe just next to each other, if the amount is
- too unequal.

- Only transitions could be more improved to improve visual look
- Nothing
- AI
- Maybe a AI Assistant would be better
- A glossary oft the symbols used for identifying what rights are impacted by a given policy could be useful, to reference when the meaning is not immediately clear.
- There are missing white spaces in the demo
- make comparison side by side fit to window, scrollbar from left to right is missing and moving the page by dragging / generally having to scroll left to right while reading is user unfriendly
- there is room for improvement with integrating AI
- Minor bug fixes (if any)

References

[1] Open Terms Archive. *Open Terms Archive. Tracking Changes in Online Terms and Policies.* Non-profit initiative providing an open infrastructure for collecting and versioning online terms of service and privacy policies. 2025. URL: `https://opentermsarchive.org/` (visited on 01/13/2026).

[2] Open Terms Archive. *Collections of Open Terms Archive.* GitHub repository of the Open Terms Archive. 2025. URL: `https://github.com/OpenTermsArchive` (visited on 01/15/2025).

[3] Chakra UI Team. *Chakra UI.* `https://www.chakra-ui.com/`. A simple, modular, and accessible component library for React applications.

[4] i18next Team. *React i18next.* `https://react.i18next.com/`. Internationalization framework for React apps, enabling multi-language support.

[5] kpdecker. *jsdiff.* `https://github.com/kpdecker/jsdiff#readme`. JavaScript library for computing differences between text snippets, useful for change tracking.

[6] React Team. *React + TypeScript Documentation.* `https://react.dev/learn/typescript`. Official guide for integrating TypeScript with React projects.

[7] React Team. *React Official Website.* `https://react.dev/`. Core library for building interactive user interfaces.

[8] RemarkJS Team. *React Markdown (RemarkJS).* `https://github.com/remarkjs/react-markdown#readme`. Library to render Markdown content as React components.